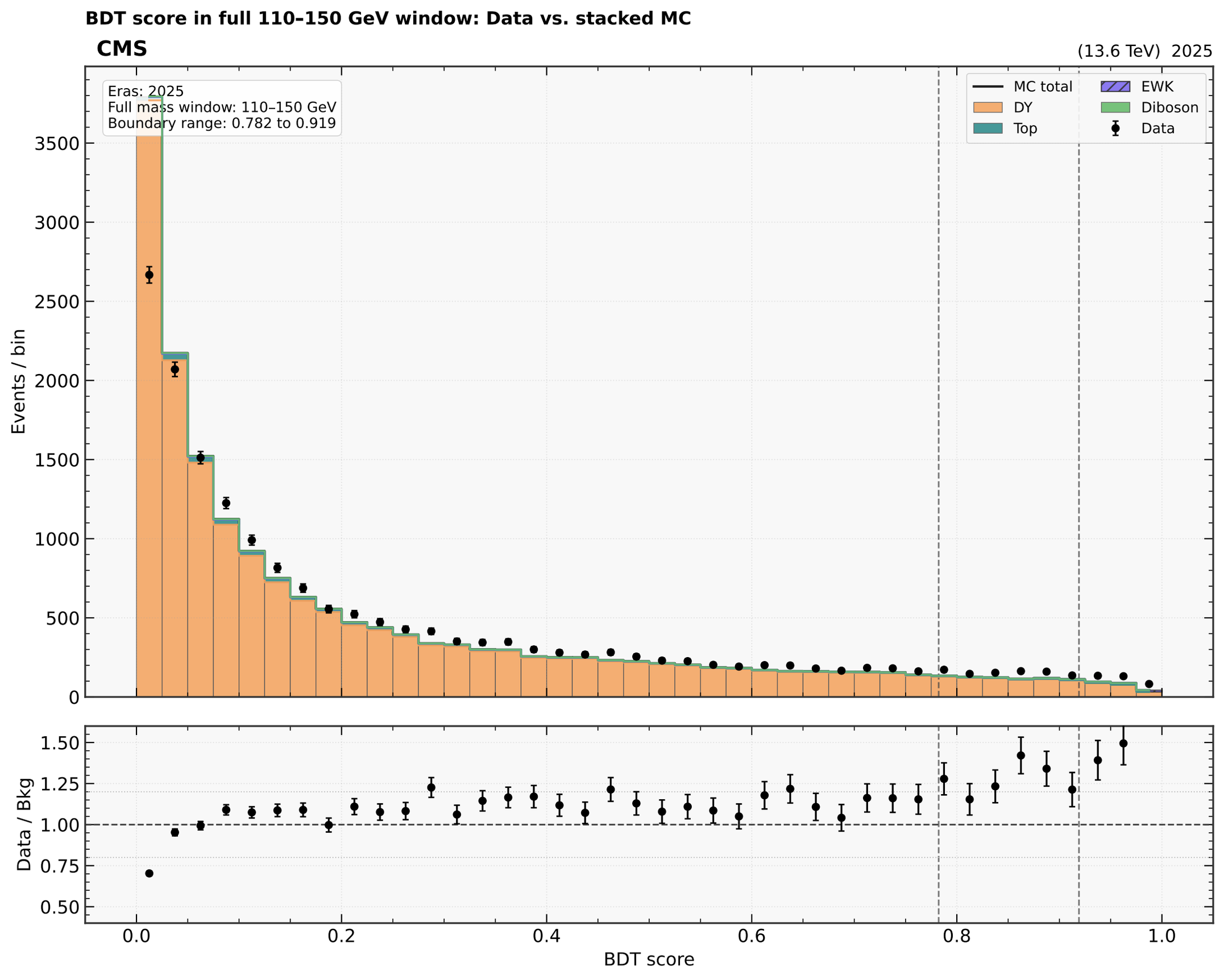
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 2025
Category edges : 0.0000, 0.7823, 0.9191, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ^2/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_{CB} vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

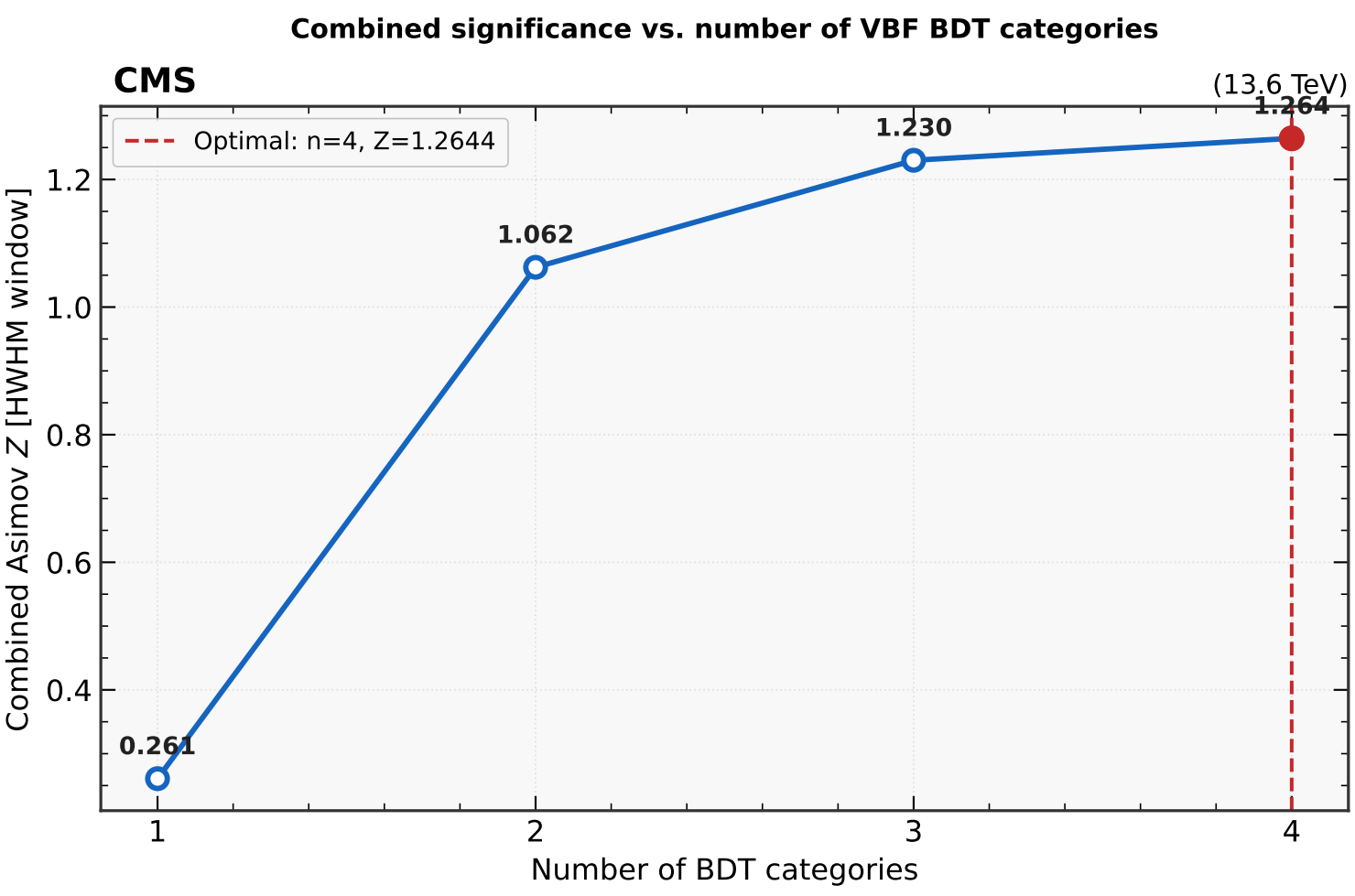

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 2025
Eras: 2025
Full mass window: 110–150 GeV
Boundary range: 0.782 to 0.919
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

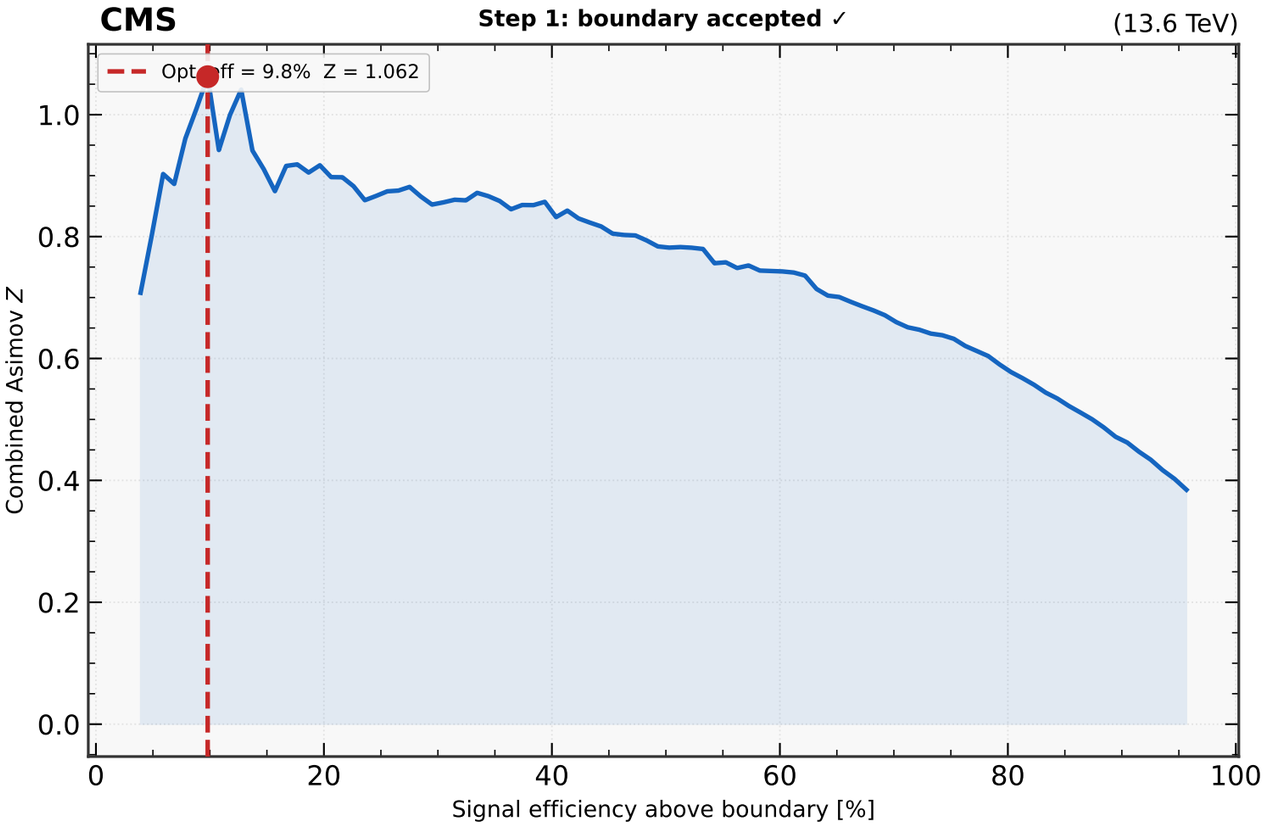
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.104	3794.56 ± 35.40	2667	0.00
[0.025, 0.050)	0.193	2172.80 ± 25.67	2070	0.01
[0.050, 0.075)	0.210	1521.84 ± 21.06	1512	0.01
[0.075, 0.100)	0.219	1123.92 ± 17.88	1225	0.02
[0.100, 0.125)	0.224	922.26 ± 16.40	991	0.02
[0.125, 0.150)	0.227	750.86 ± 14.70	816	0.03
[0.150, 0.175)	0.237	631.34 ± 13.21	688	0.04
[0.175, 0.200)	0.233	556.42 ± 12.36	555	0.04
[0.200, 0.225)	0.246	471.32 ± 11.26	523	0.05
[0.225, 0.250)	0.251	439.41 ± 11.11	473	0.06
[0.250, 0.275)	0.256	394.47 ± 10.53	427	0.06
[0.275, 0.300)	0.262	338.48 ± 9.59	415	0.08
[0.300, 0.325)	0.264	330.67 ± 9.46	351	0.08
[0.325, 0.350)	0.280	300.50 ± 9.19	344	0.09
[0.350, 0.375)	0.296	298.61 ± 9.03	348	0.10
[0.375, 0.400)	0.296	256.29 ± 8.25	300	0.12
[0.400, 0.425)	0.303	250.50 ± 8.15	280	0.12
[0.425, 0.450)	0.333	250.07 ± 8.46	268	0.13
[0.450, 0.475)	0.350	232.28 ± 7.85	282	0.15
[0.475, 0.500)	0.361	225.79 ± 7.90	255	0.16

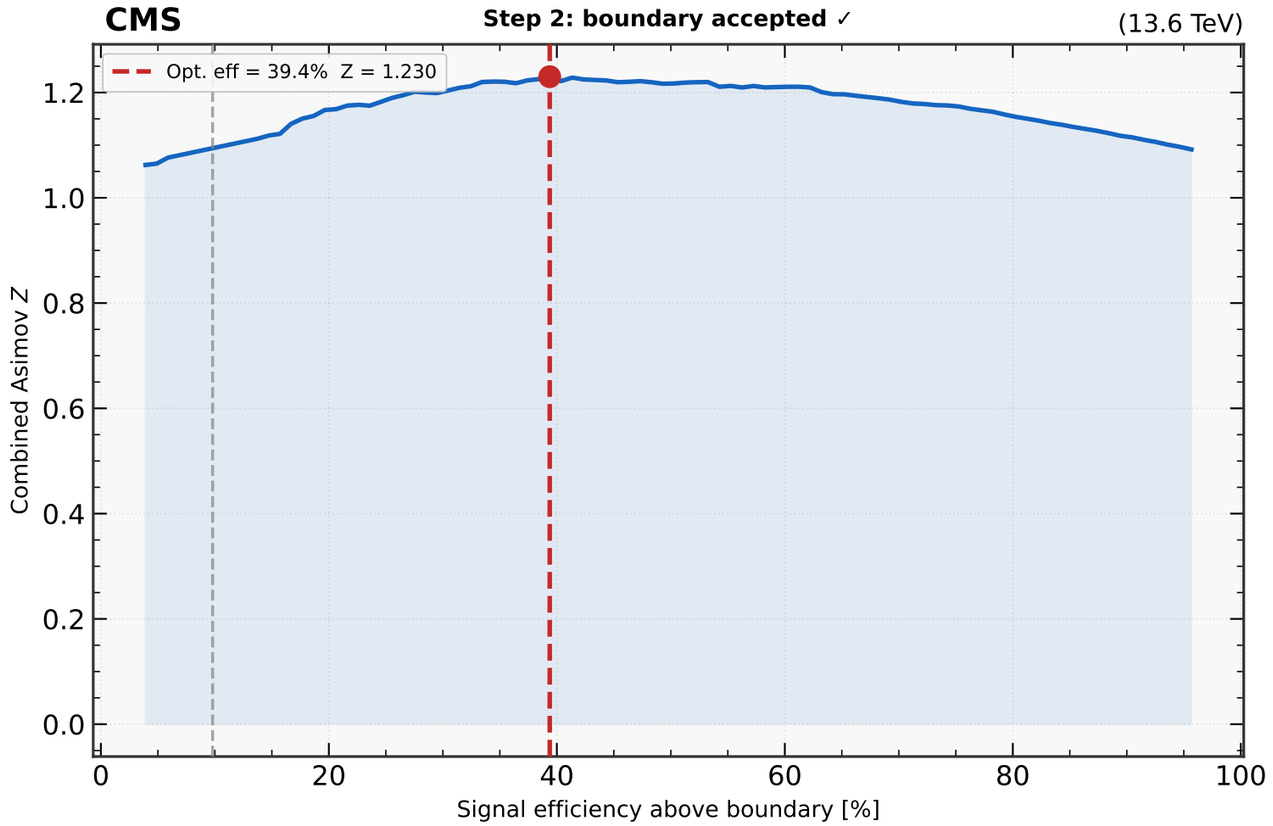
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

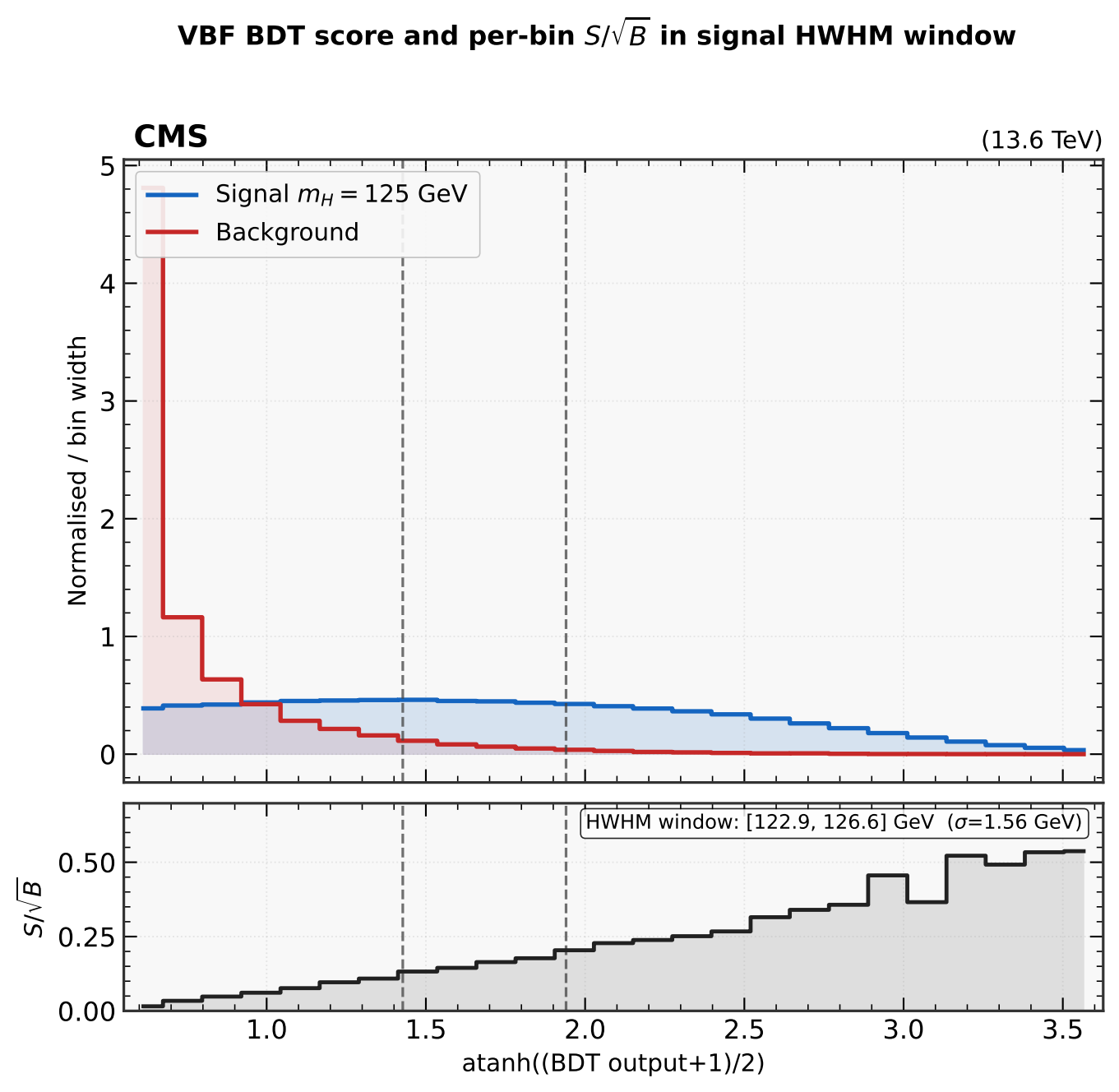
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.386	213.15 ± 7.71	230	0.18
[0.525, 0.550)	0.391	203.74 ± 7.71	226	0.19
[0.550, 0.575)	0.426	186.99 ± 7.10	203	0.23
[0.575, 0.600)	0.450	182.86 ± 6.95	192	0.25
[0.600, 0.625)	0.478	170.55 ± 6.74	201	0.28
[0.625, 0.650)	0.515	163.40 ± 6.47	199	0.31
[0.650, 0.675)	0.542	162.48 ± 6.53	180	0.33
[0.675, 0.700)	0.591	159.38 ± 6.46	166	0.37
[0.700, 0.725)	0.636	158.30 ± 6.35	184	0.40
[0.725, 0.750)	0.705	155.91 ± 6.36	181	0.45
[0.750, 0.775)	0.756	140.38 ± 5.90	162	0.54
[0.775, 0.800)	0.861	134.51 ± 5.91	172	0.64
[0.800, 0.825)	0.949	126.52 ± 5.62	146	0.74
[0.825, 0.850)	1.076	124.08 ± 5.60	153	0.86
[0.850, 0.875)	1.262	114.70 ± 5.24	163	1.09
[0.875, 0.900)	1.514	119.36 ± 5.49	160	1.25
[0.900, 0.925)	1.878	112.09 ± 5.19	136	1.65
[0.925, 0.950)	2.500	96.25 ± 4.79	134	2.53
[0.950, 0.975)	3.713	87.64 ± 4.31	131	4.06
[0.975, 1.000)	5.024	42.96 ± 2.92	82	10.47

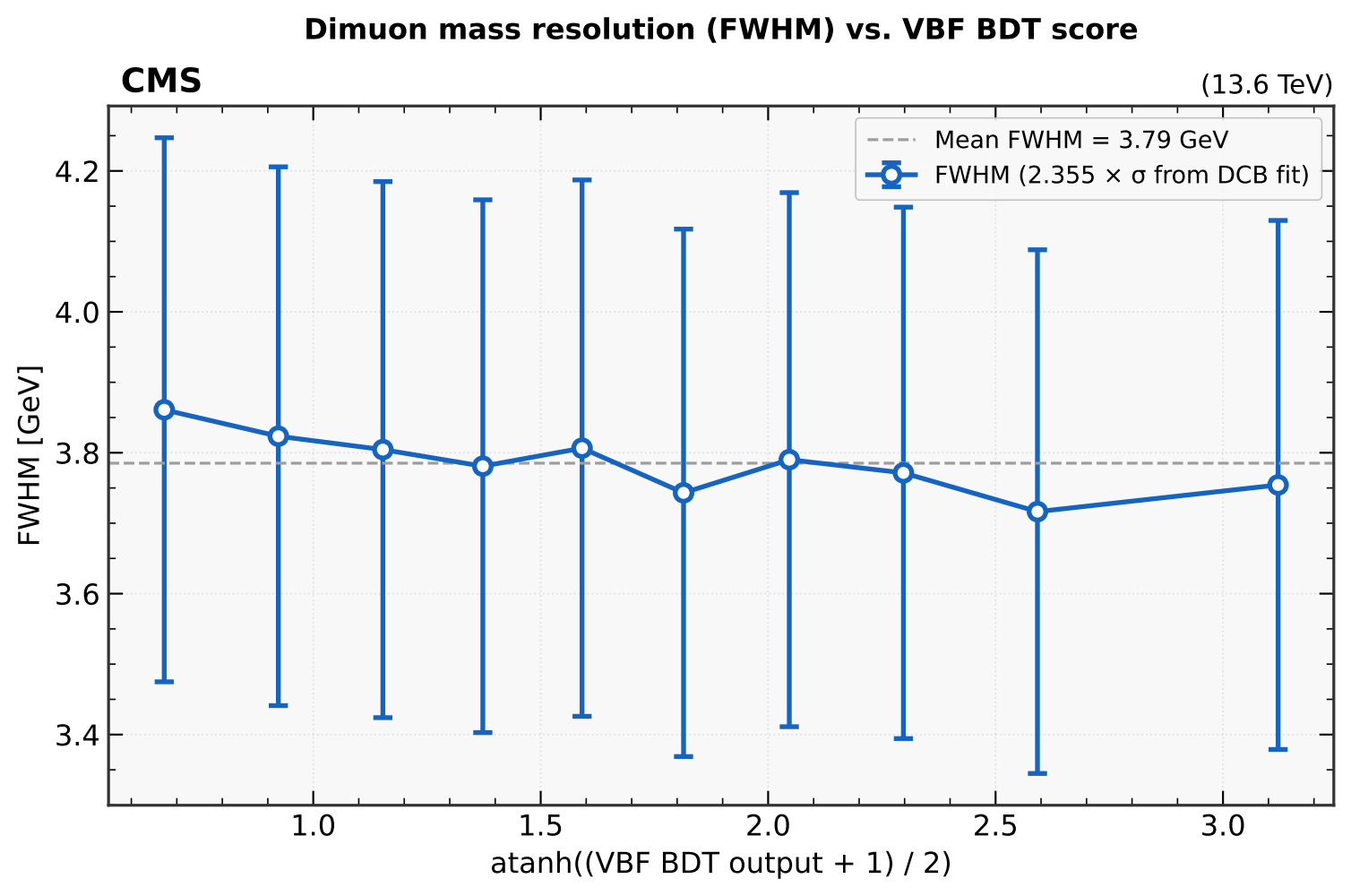

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.2644
0.261
1.062
1.230
1.264
Combined Asimov Z [HWHM window]
Number of BDT categories
0.4
0.6
0.8
1.0
1.2
1
2
3
4

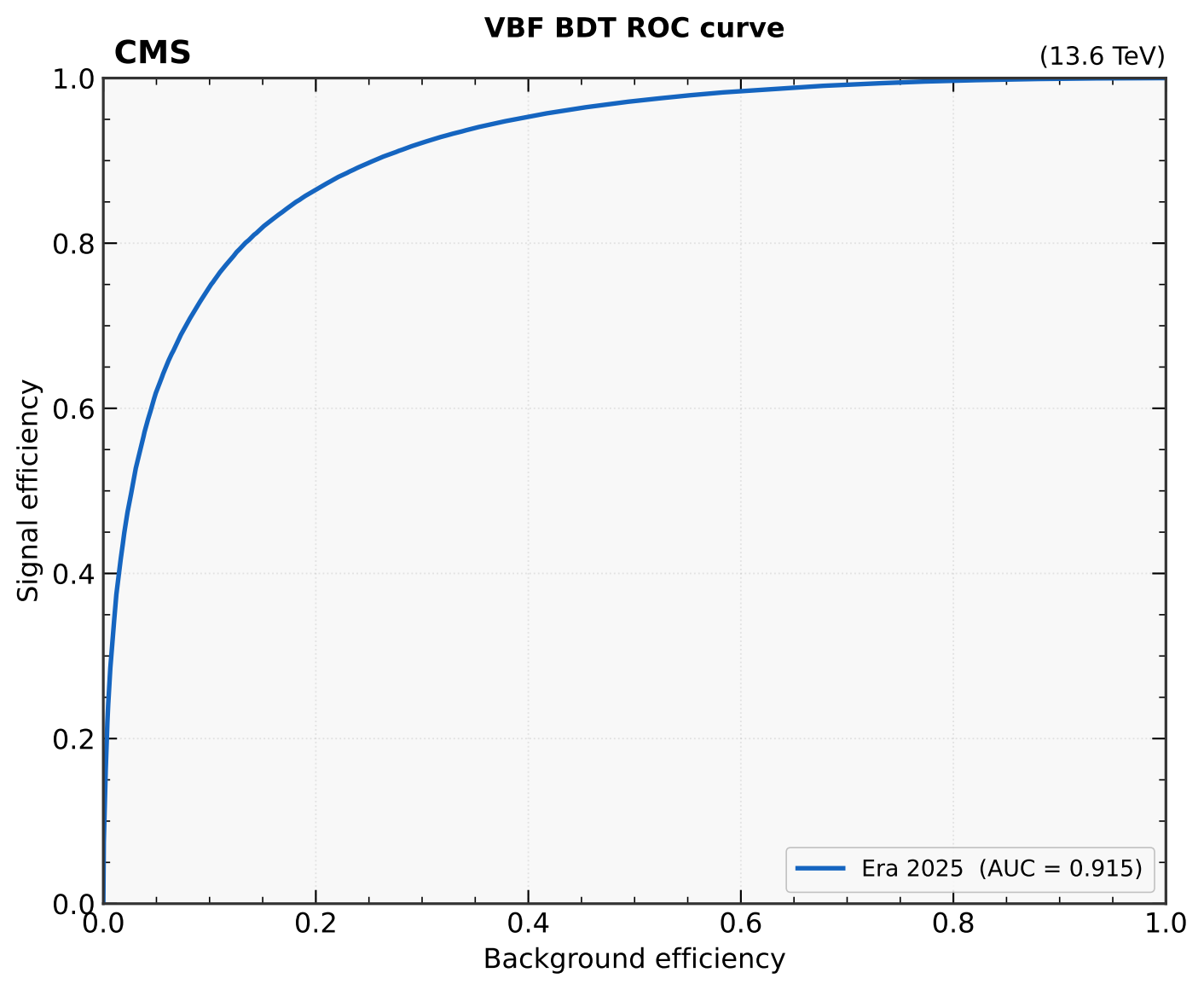
Iterative category boundary optimisation (AN Fig. 32 analog)

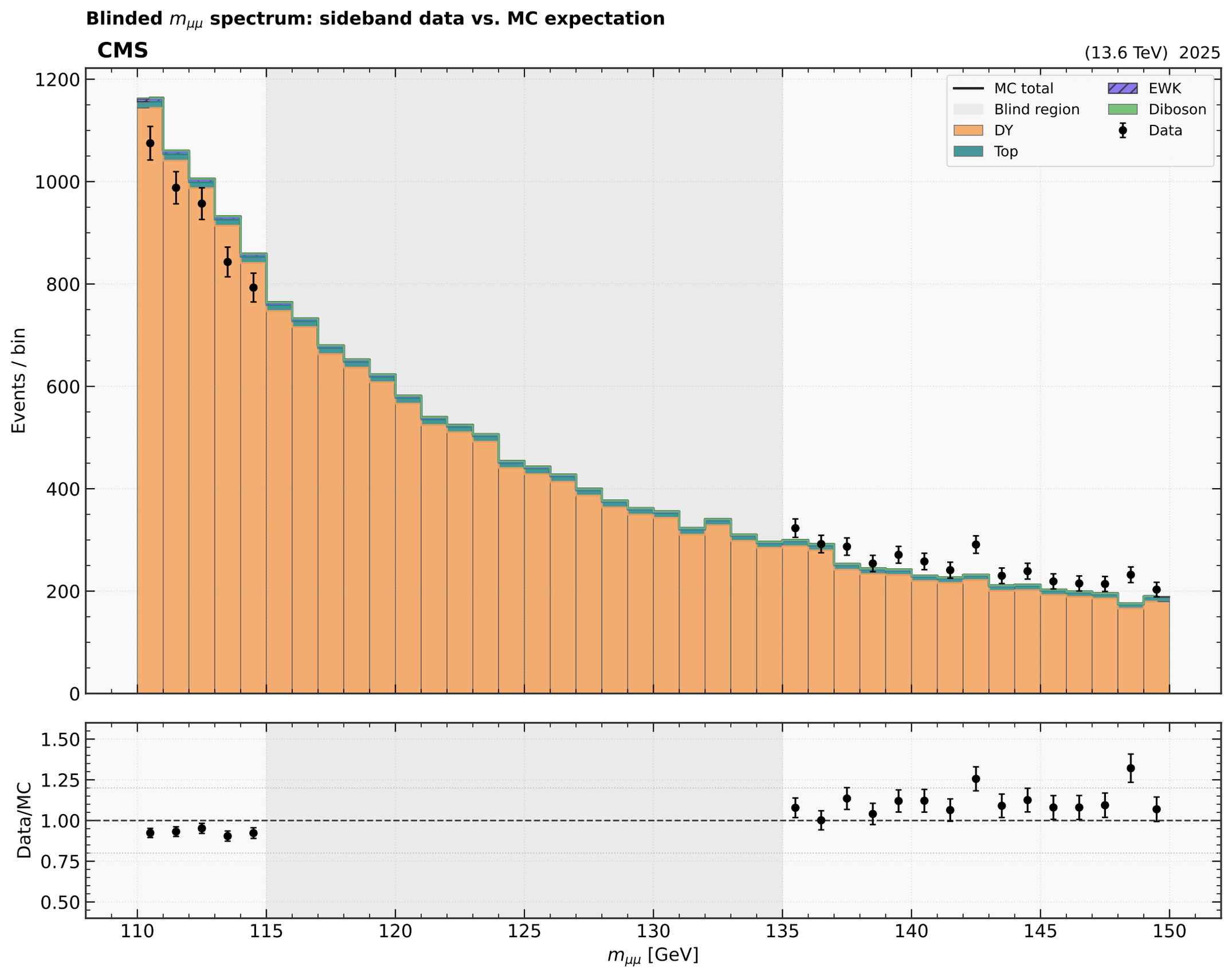


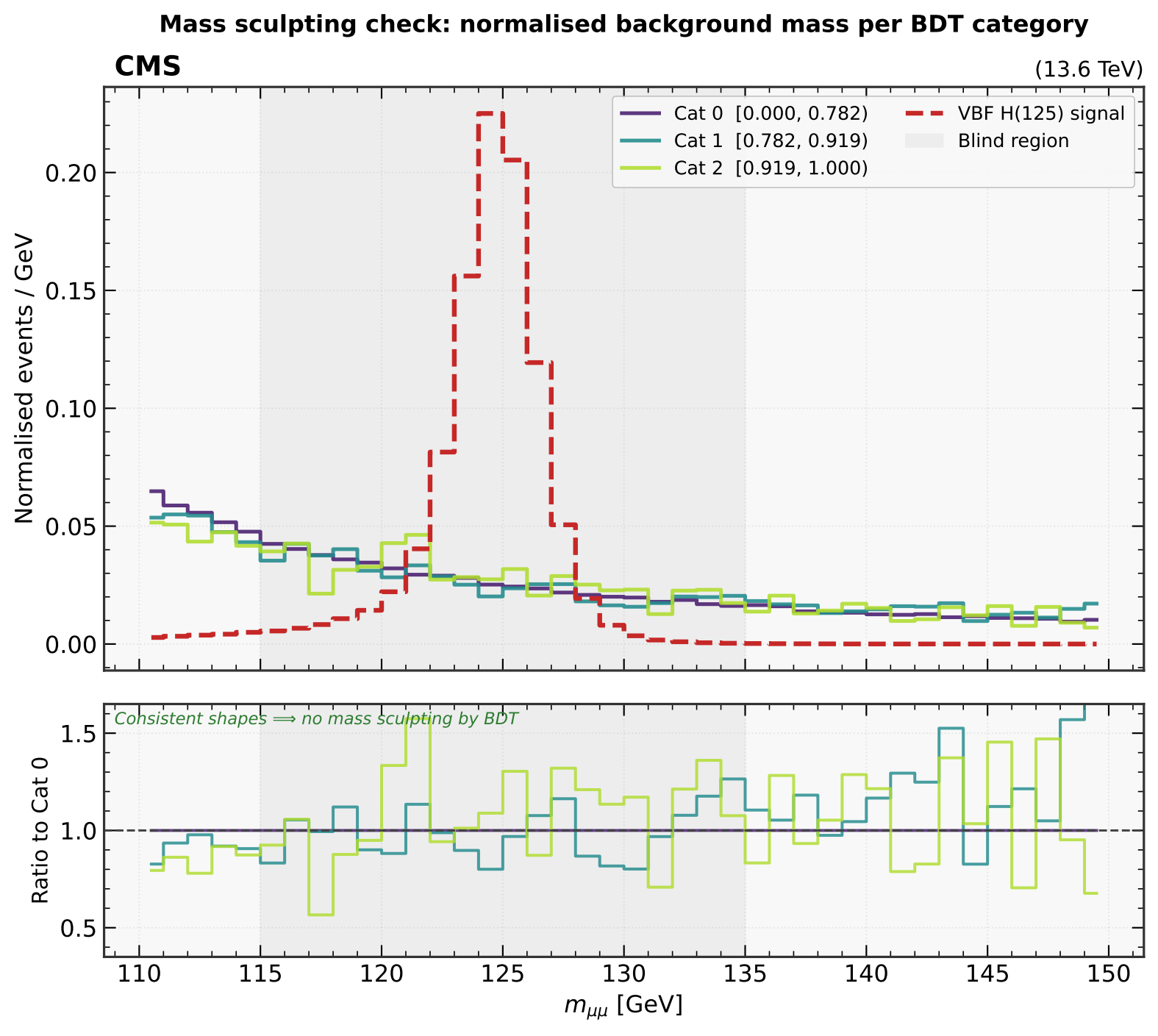



VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window
CMS
(13.6 TeV)
Signal $m_H = 125$ GeV
Background
Normalised / bin width
0
1
2
3
4
5
HWHM window: [122.9, 126.6] GeV (σ=1.56 GeV)
$S/\sqrt{B}$
0.00
0.25
0.50
1.0
1.5
2.0
2.5
3.0
3.5
atanh((BDT output+1)/2)

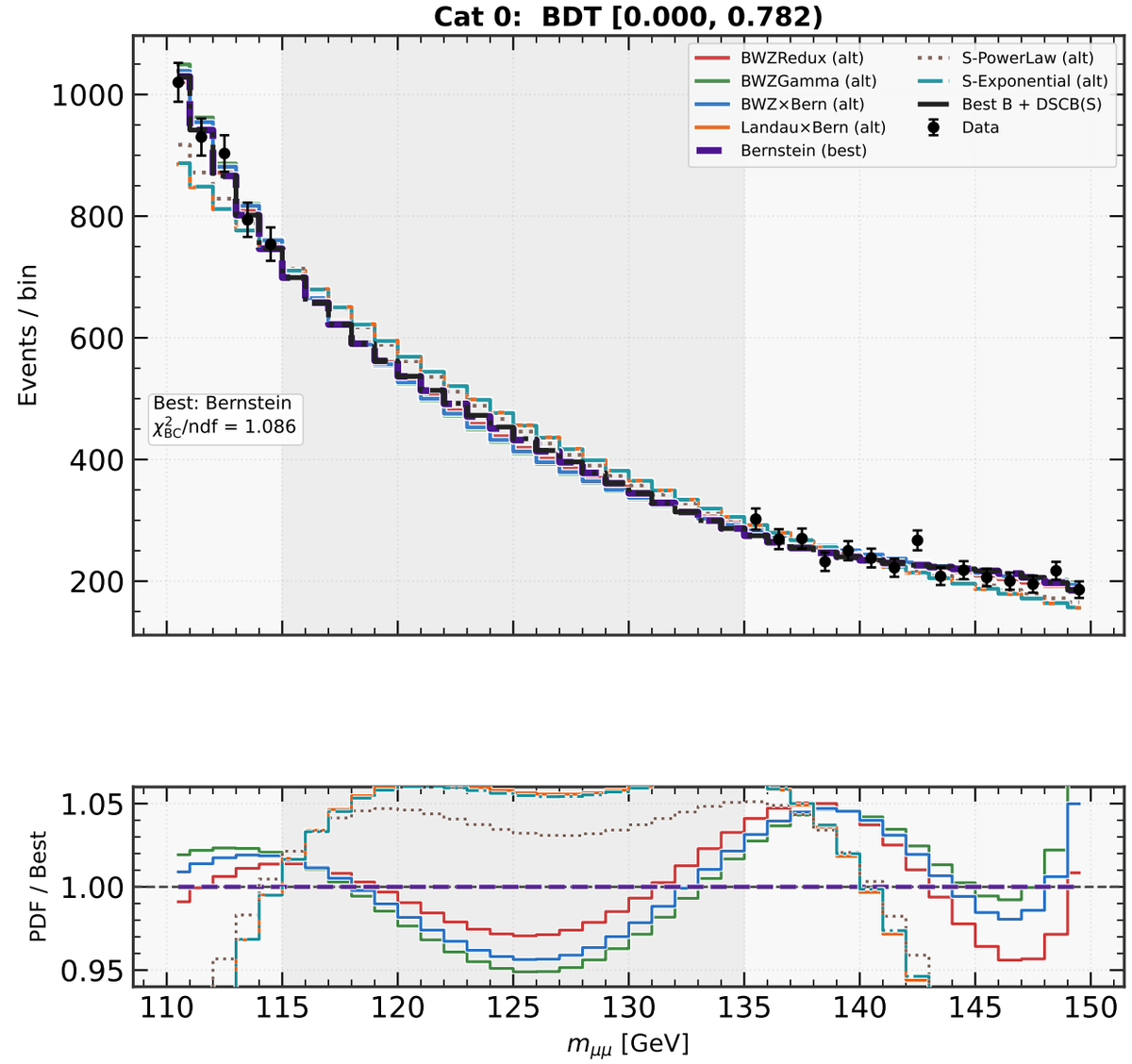

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.79 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

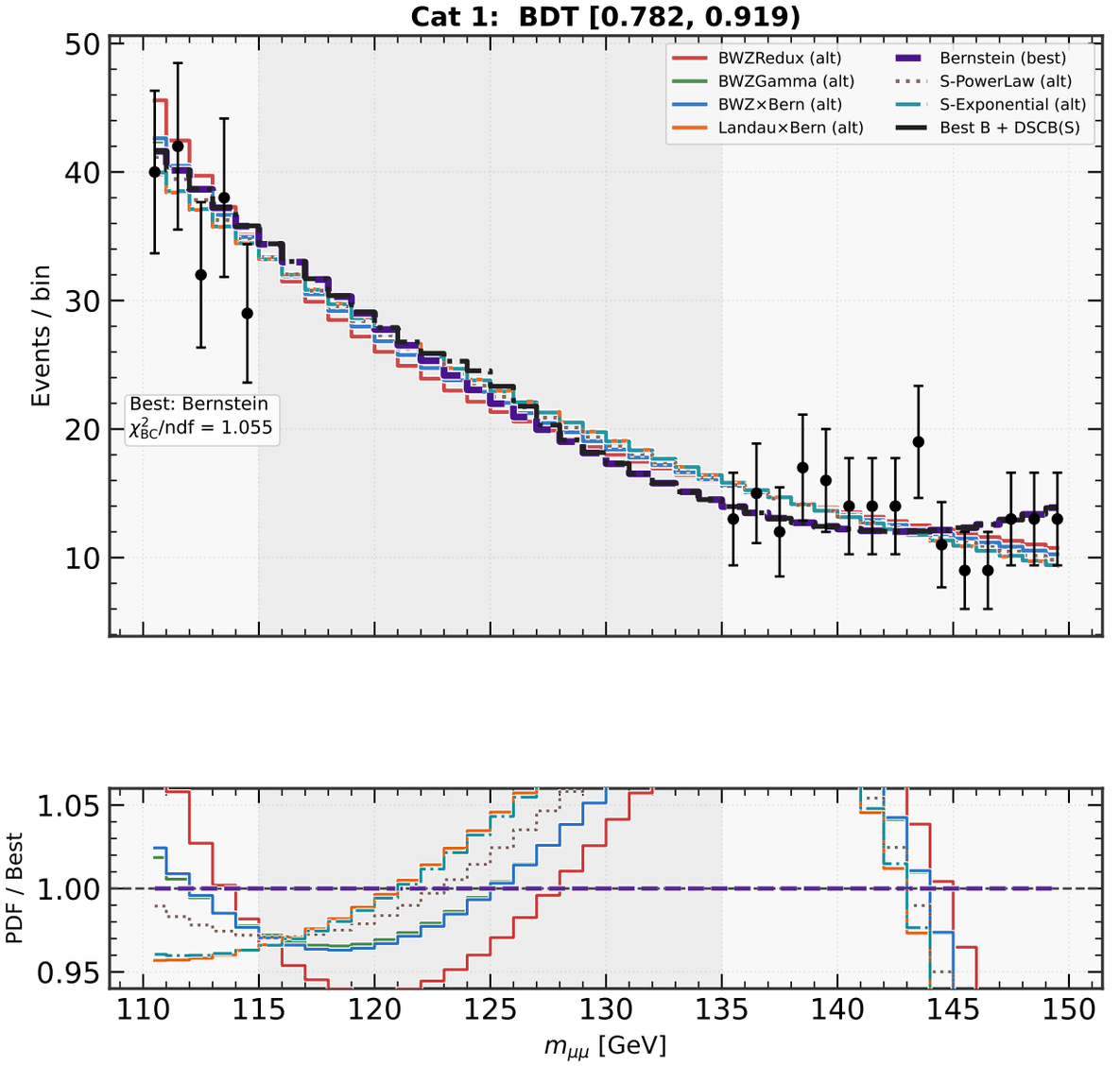

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 2025 (AUC = 0.915)

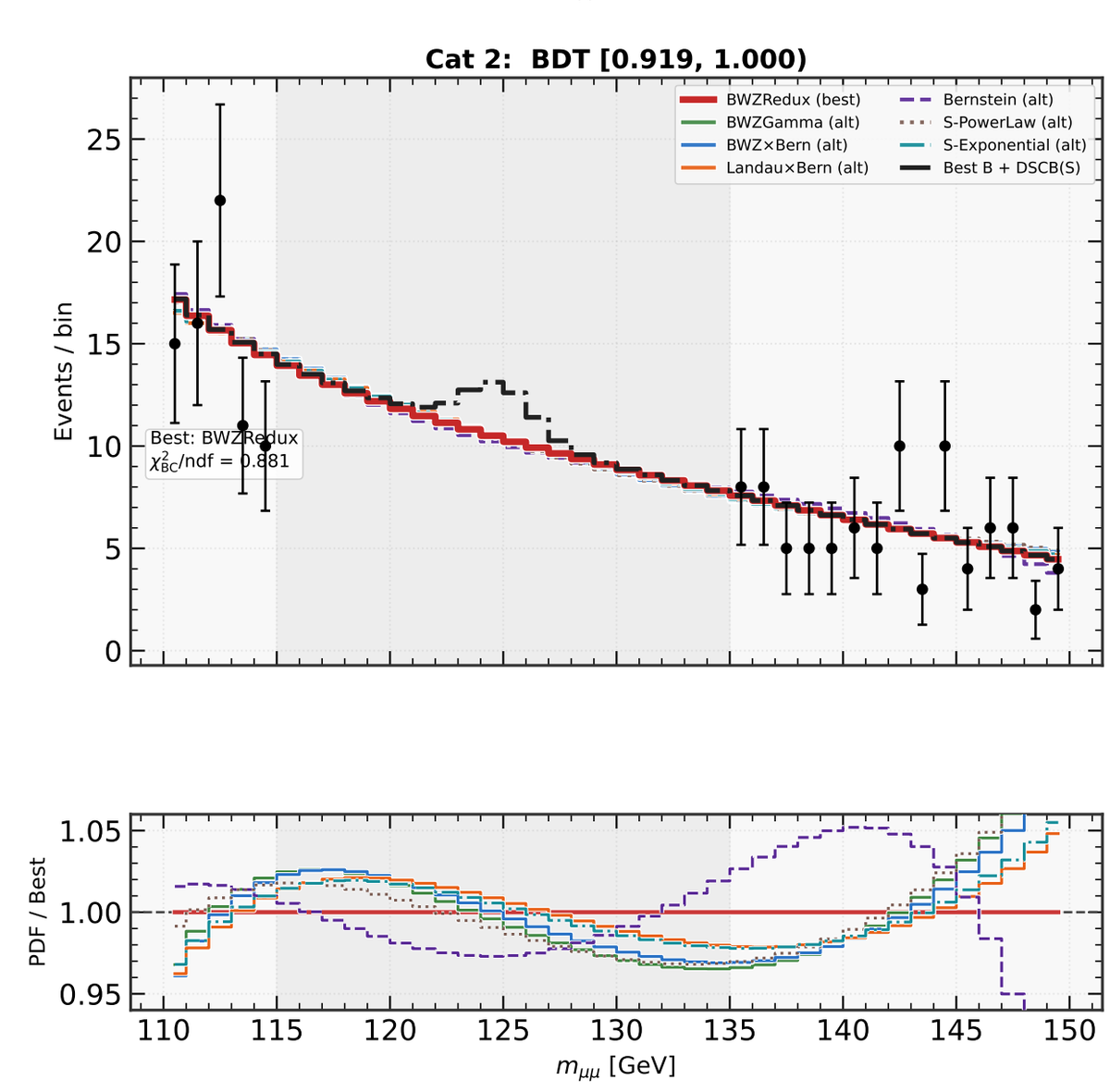

Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 2025
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]


Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.782)
Cat 1 [0.782, 0.919)
Cat 2 [0.919, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits



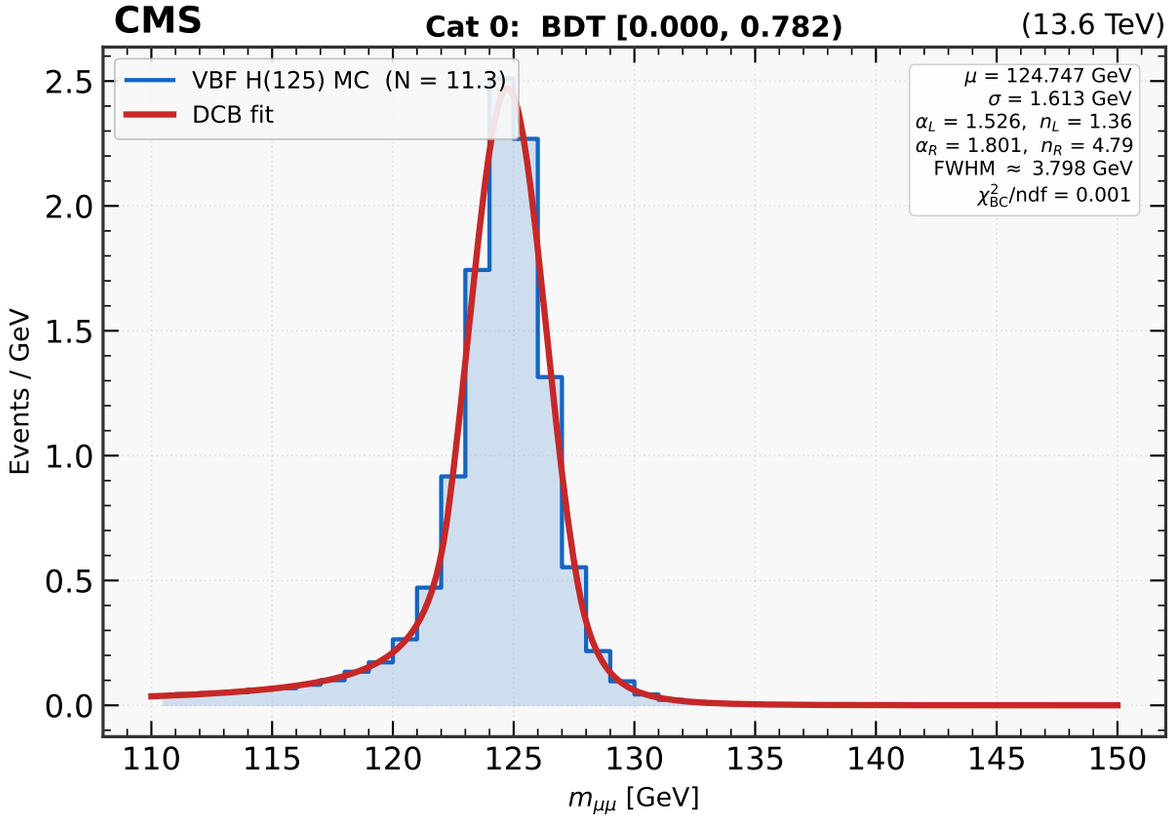


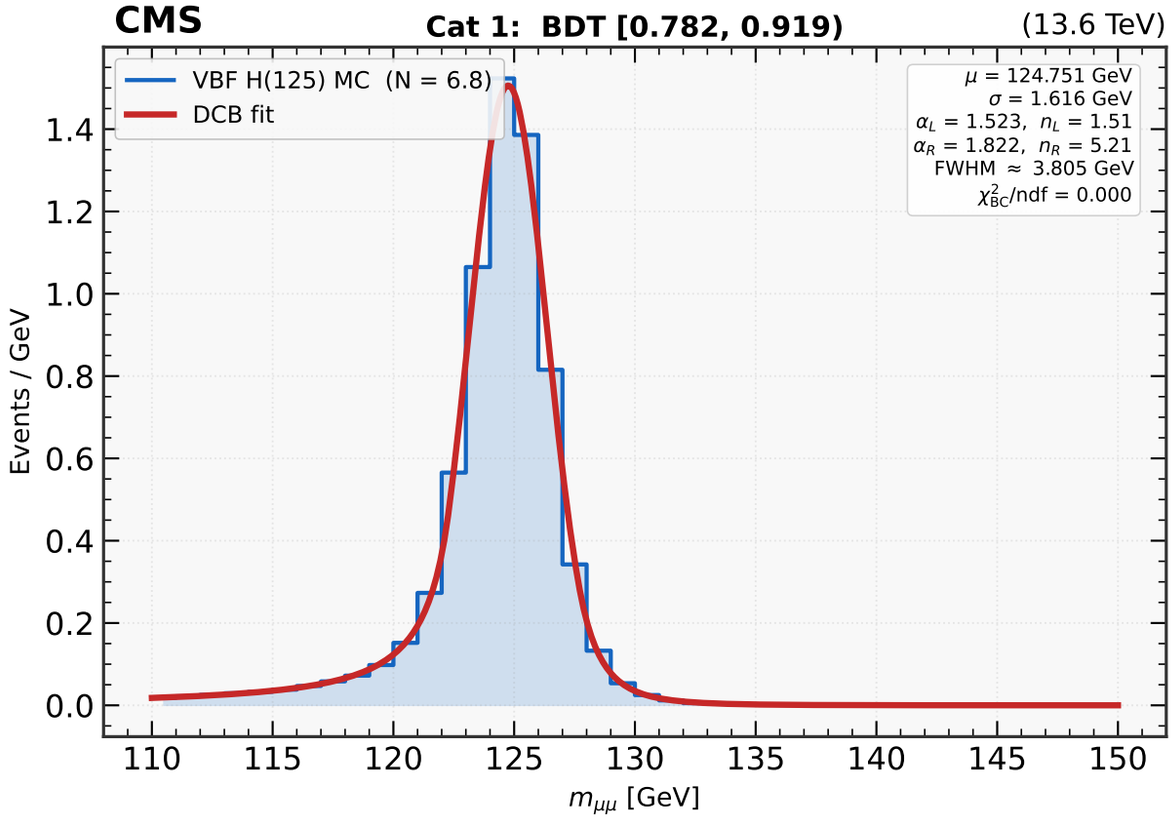


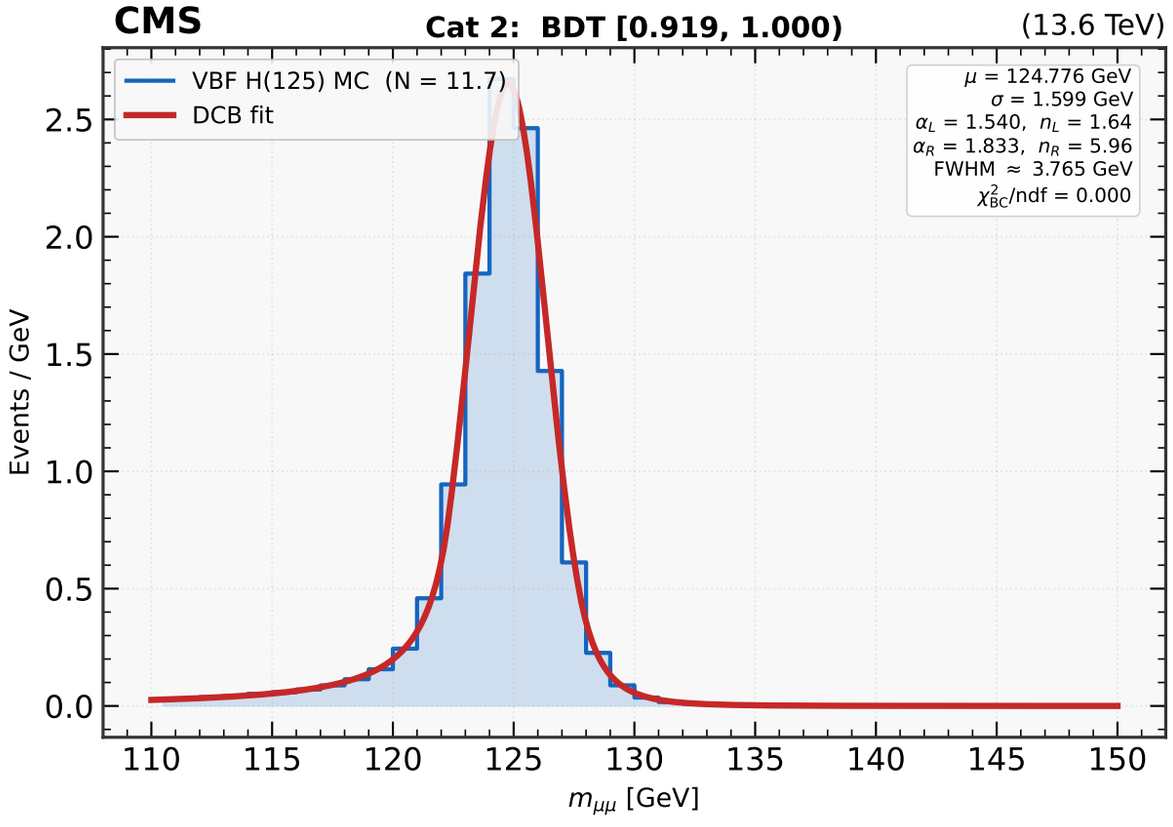
Per-category background model fit comparison (χ²_BC/ndf, ndf)

Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.782)	Bernstein	1.086	1.146 (37)	1.325 (37)	1.255 (36)	3.702 (37)	1.086 (33)	2.465 (37)	3.598 (37)
1	[0.782, 0.919)	Bernstein	1.055	1.166 (37)	1.125 (37)	1.163 (36)	1.187 (37)	1.055 (35)	1.145 (37)	1.183 (37)
2	[0.919, 1.000)	BWZRedux	0.881	0.881 (37)	0.889 (37)	0.913 (36)	0.887 (37)	0.917 (35)	0.889 (37)	0.886 (37)

VBF H(125) signal DCB extraction per BDT category

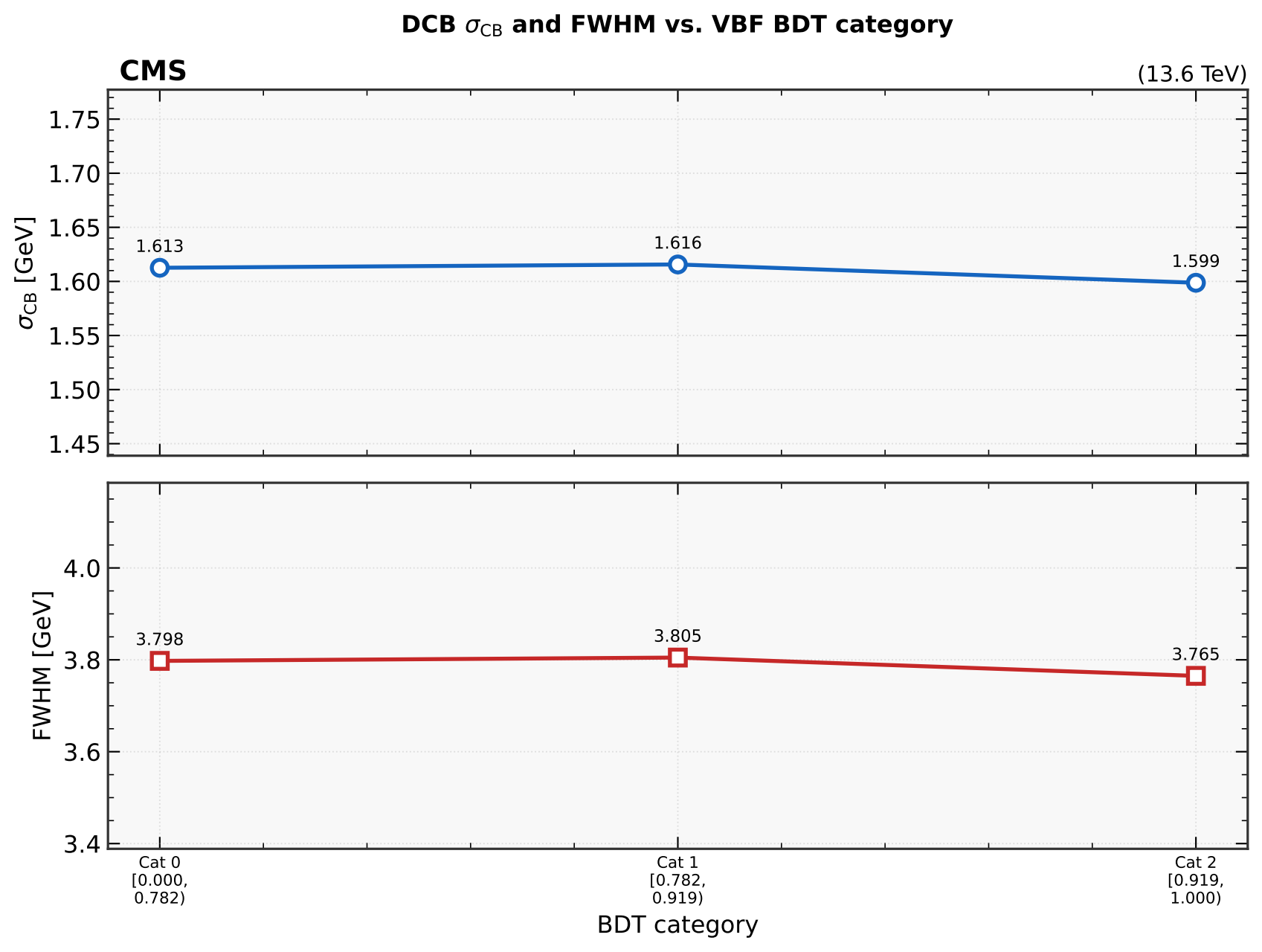


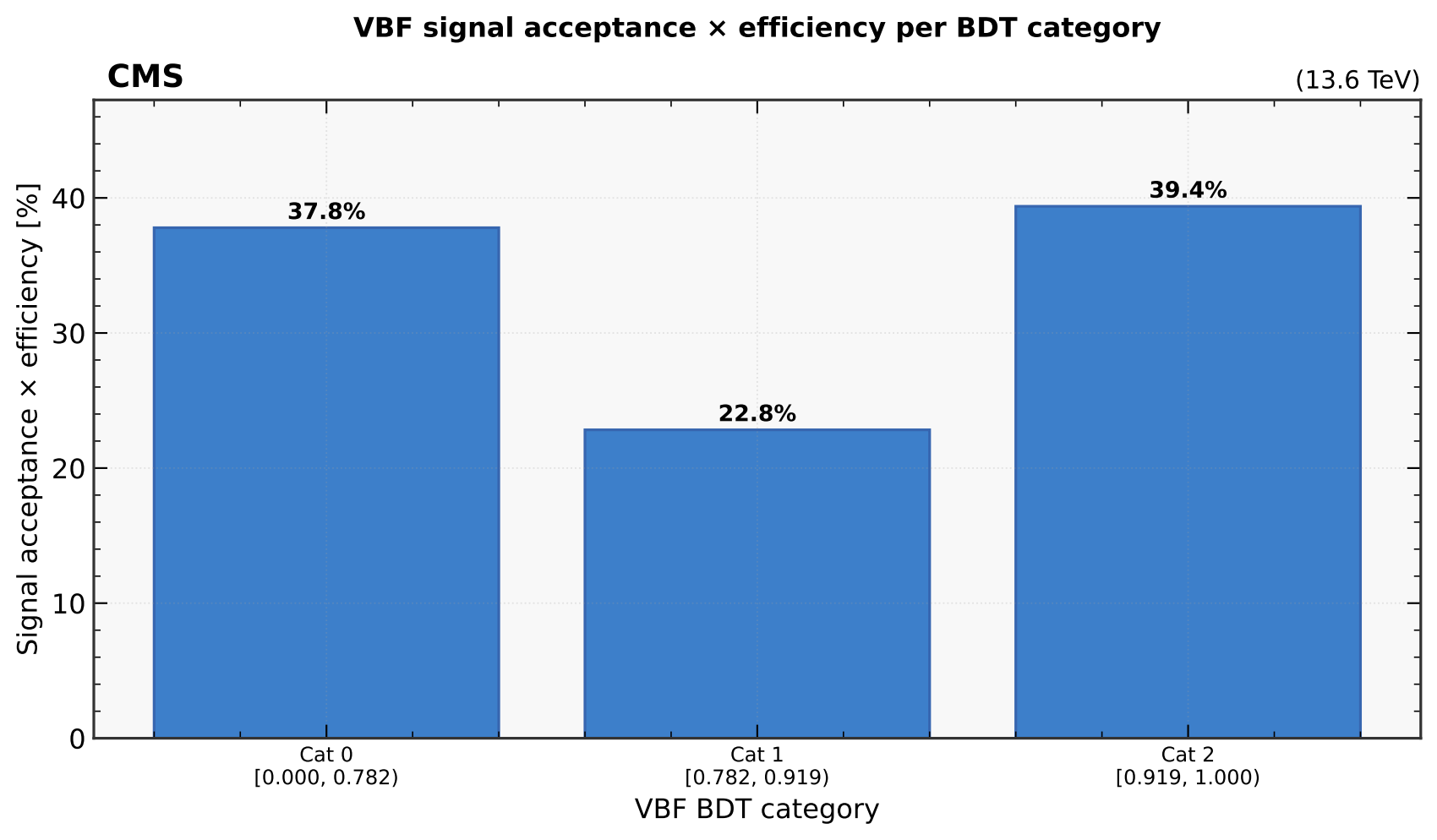




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.782)	11.26	124.7466	1.6126	1.5258	1.358	1.8013	4.790	3.798	0.100	0.001
1	[0.782, 0.919)	6.80	124.7515	1.6157	1.5233	1.512	1.8221	5.213	3.805	0.316	0.000
2	[0.919, 1.000)	11.73	124.7755	1.5988	1.5400	1.638	1.8335	5.957	3.765	0.840	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.75
1.70
1.65
1.60
1.55
1.50
1.45
1.613
1.616
1.599
FWHM [GeV]
4.0
3.8
3.6
3.4
3.798
3.805
3.765
Cat 0
[0.000,
0.782)
Cat 1
[0.782,
0.919)
Cat 2
[0.919,
1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
37.8%
22.8%
39.4%
Cat 0
[0.000, 0.782)
Cat 1
[0.782, 0.919)
Cat 2
[0.919, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.782)	11.26	59784.13	7.662	5853.044	[122.8, 126.6]	0.100	0.100
1	[0.782, 0.919)	6.80	2302.51	4.684	217.922	[122.8, 126.7]	0.317	0.316
2	[0.919, 1.000)	11.73	875.10	8.169	91.982	[122.9, 126.7]	0.852	0.840

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.782)	7881	27806.36 ± 168.14	0.283	59784.13 ± 246.78	0.1002	0.1001	5853.044
1	[0.782, 0.919)	383	1093.69 ± 30.83	0.350	2302.51 ± 44.61	0.3173	0.3162	217.922
2	[0.919, 1.000)	161	378.15 ± 16.59	0.426	875.10 ± 26.03	0.8518	0.8396	91.982
Combined	all						0.9027	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.782)	58164.46	155.16	1321.84	142.66	59784.13	60900.84
1	[0.782, 0.919)	2171.63	72.79	50.07	8.02	2302.51	1990.49
2	[0.919, 1.000)	750.84	102.76	18.62	2.88	875.10	677.58